LiveBetter | Hackathon Decision Log

Team: Athena Theme: Powered to Live Better Hackathon: MDX IET Life Powered by Tech
Period: 13 November 2026, 1:00 PM - 14 November 2026, 9:00 AM Time zone: MUT (UTC+4)

Date & time	Key decision	Reason / outcome
13 Nov 1:15 PM	**1. Selected an integrated household-management solution**	We chose to address three connected challenges faced by Mauritian households: utility expenses, grocery management and food waste. LiveBetter would handle them in one application rather than three separate solutions.
13 Nov 2:00 PM	**2. Divided the work across the project components**	We divided development across three main modules - Water, Electricity and Food - and a central AI Coach that collects and combines insights from all three. Mobile application and integration work connected these components into one experience.
13 Nov 3:10 PM	**3. Established a unified cloud-based architecture**	We selected React Native with Expo, Python FastAPI, Supabase PostgreSQL and Render. A shared backend and household identity would connect the modules and support mobile deployment.
13 Nov 4:30 PM	**4. Chose explainable AI with verifiable calculations**	We decided that AI-generated explanations must be grounded in values calculated by the modules. The AI Coach would validate, prioritise and explain recommendations across Water, Electricity and Food rather than inventing savings figures.
13 Nov 6:15 PM	**5. Grounded the solution in the Mauritian context**	We focused on Mauritian households by incorporating CWA and CEB tariff rules, local temperature data and relevant real-world water-consumption research. Household bills and meter readings personalise forecasts, while English, French and Kreol Morisien improve local accessibility.
13 Nov 8:00 PM	**6. Responded to the surprise card: MISUSE**	After drawing MISUSE, we chose server-verified security patterns for sensitive actions, Microsoft Authenticator as an optional second factor, failed-attempt lockouts, security alerts, activity logs and a 24-hour reset recovery option.
13 Nov 9:40 PM	**7. Addressed privacy concerns in document processing**	We compared traditional OCR, which typically extracts all page text, with LLM-based extraction of only the required bill or receipt fields. We chose the LLM approach to minimise personal information retained in the database, subject to provider-side processing and user confirmation before saving.

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14 Nov 12:30 AM	**8. Adopted a rule-based AI Coach for actionable guidance**	We chose rule-based logic for household what-if calculations, with the language model assisting conversation and explanations. The AI Coach combines module insights, asks follow-up questions when inputs are missing and applies accepted changes to forecasts without allowing the model to invent numbers.
14 Nov 4:15 AM	**9. Prioritised accessibility and understandable feedback**	We incorporated English, French and Kreol Morisien, straightforward navigation, clear error messages and PDF-exportable activity records so households can understand their data and recommendations.
14 Nov 9:00 AM	**10. Completed the final testing cycle**	We completed a final testing cycle covering the three modules, AI Coach, mobile-to-backend integration, data flows and MISUSE safeguards. We also prepared fallback demonstration options for connectivity or deployment issues.

Final outcome

LiveBetter combines household expense forecasting, food inventory management and a rule-based, explainable AI Coach within one application. It helps Mauritian families understand costs, use pantry ingredients effectively, evaluate daily choices and protect sensitive household information.